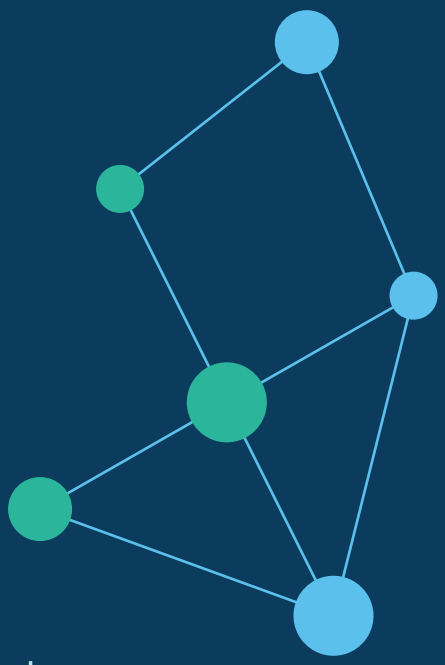

Evaluation Report

AI-Powered Telecom Network Operations Assistant

How the assistant was evaluated and what the results show

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Table of Contents

1. Objectives and Scope 3

2. Evaluation Approach 3

3. Metrics Used 3

3.1 DeepEval metrics . . . 3

3.2 RAGAS metrics . . . 3

4. LLM-as-Judge Setup 4

5. Results 4

5.1 ML model evaluation . . . 4

5.2 RAG retrieval quality . . . 4

5.3 DeepEval test case: latency_above_130 . . . 4

6. Thresholds and Result Labels 4

7. Findings by Area 5

8. Limitations 5

9. Recommendations 5

10. Source Files and References 5

1. Objectives and Scope

This report explains how the assistant is evaluated and presents the results that were available when it was written. The goal is to check four things: whether retrieval finds relevant and filter-compliant incidents, whether RCA output is faithful and cautious, whether the final answer is relevant and grounded, and whether the prediction models fit held-out data.

> Scope of results: the full evaluation JSON reports (evaluations/reports/) were not part of the material used to write this report. It contains the ML model metrics, the overall RAG retrieval quality score, and one DeepEval test case (latency_above_130). Anything else is marked as not available. No scores were estimated or filled in.

2. Evaluation Approach

Method	What it checks	Tool
Agent and answer evaluation	Retrieval, RCA and final-answer quality using an LLM judge	DeepEval
RAG evaluation	Faithfulness, context precision and context relevance for the Synapt / RAG path	RAGAS
Model evaluation	R-squared, MAE and RMSE on held-out test data	scikit-learn metrics
Contract checks	Prediction-agent output contract and focused tests	Test suite
Feedback evaluation	Helpful / not-helpful ratings stored for later review	SQLite (feedback/feedback.sqlite3)

Reports are saved as JSON in evaluations/reports/, and the Evaluations page in the React frontend reads the saved report data.

3. Metrics Used

3.1 DeepEval metrics

Stage	Metric	Meaning
Retrieval	Contextual Precision	Relevant incidents are ranked above less relevant ones.
Retrieval	Contextual Recall	The retrieved context contains the information needed to answer.
Retrieval	Contextual Relevancy	The retrieved context is relevant to the question.
Retrieval	Retrieval Filter Compliance	Every returned row satisfies the requested filters.
RCA	RCA Quality (GEval)	Hypotheses are ranked, use the retrieved context, state uncertainty and name the symptom.
RCA	Faithfulness	The RCA output is consistent with the retrieved context.
RCA	RCA Evidence and Caution	Hypotheses are evidence-linked and the wording stays cautious.
Final answer	Answer Relevancy	The answer addresses the request.
Final answer	Faithfulness	The answer is consistent with the retrieved context.
Final answer	Hallucination	The answer does not add unsupported content.

3.2 RAGAS metrics

Metric	Meaning
Faithfulness	Answer statements are supported by the retrieved context.
Context Precision	How much of the retrieved context is relevant, and how well relevant items are ranked.
Context Relevance	How relevant the retrieved context is to the question.

4. LLM-as-Judge Setup

Item	Setting
Provider	AICredits (OpenAI-compatible API)
Judge model	google/gemini-3.1-flash-lite (set through DEEPEVAL_MODEL and the AICredits model settings)
Client	llm/llm_client.py: provider selection, timeout, temperature, token limit and retry settings
Timeout and retry values	Configured in the project. The exact values are not in the material used for this report.

5. Results

5.1 ML model evaluation

Trained on 80,000 samples and evaluated on 20,000 held-out test samples (regression metrics).

Model	R-squared	MAE	RMSE	Status
Network performance model (throughput)	0.9782	29.44 Mbps	35.42 Mbps	good
Fault risk model (fault occurrence rate)	0.9907	1.07 %	1.35 %	good

5.2 RAG retrieval quality

Overall RAG retrieval quality: **0.96**.

5.3 DeepEval test case: latency_above_130

Query type: historical incidents with latency above 130 ms. This is the only per-query result available for this report.

Stage	Metric	Score	Result	Reason (summary)
Retrieval	Contextual Precision	1.00	Passed	All nodes ranked correctly; each contains incidents above 130 ms.
Retrieval	Contextual Recall	1.00	Passed	Nodes 1 to 5 hold all data needed for the count and examples.
Retrieval	Contextual Relevancy	1.00	Passed	Retrieved incidents (e.g. INC058190, INC010128) meet the latency criterion.
Retrieval	Retrieval Filter Compliance	1.00	Passed	All returned rows satisfy the requested filters.
RCA	RCA Quality (GEval)	1.00	Passed	Ranks incidents, uses context, keeps appropriate uncertainty, names the latency symptom.
RCA	Faithfulness	1.00	Passed	Output consistent with the retrieval context.
RCA	RCA Evidence and Caution	1.00	Passed	Hypotheses are evidence-linked and cautious.
Final answer	Answer Relevancy	1.00	Passed	Directly addresses the request with the matching incidents.
Final answer	Faithfulness	0.86	Passed	The output stated 165 historical incidents, but the retrieval context listed only 5.
Final answer	Hallucination	1.00	Passed	No reason text recorded for this metric.

6. Thresholds and Result Labels

Pass thresholds and the labels shown in the frontend are defined in the evaluation code. The numeric cut-offs are not in the material used for this report, so none are stated here. The table below explains how each kind of result should be read (interpretation guidance).

Result	How to read it
Good	The metric met its configured threshold.

Result	How to read it
Moderate	Below the best range but not clearly failing. Review the cases behind the score.
Low	Below the acceptable range. Investigate the failing queries.
Unavailable	The metric could not be computed (no data, or the judge did not run). It is not a score of zero.
Timeout	The judge call did not return in time. It is not a quality result. Rerun the case.

7. Findings by Area

Area	Finding	Basis
Synapt retrieval (RAGAS)	Project notes describe Context Precision as moderate. Faithfulness and Context Relevance values were not provided.	Values not supplied
Complete orchestration	The example case ran retrieval, RCA and final answer and was scored on all three. Results for other queries were not provided.	One test case
Prediction	Both models score well on held-out data (R-squared 0.9782 and 0.9907).	Held-out split of the project dataset
RCA	All three RCA metrics scored 1.00 on the example.	One test case
Faithfulness	RCA scored 1.00. Final-answer faithfulness scored 0.86 because the answer stated 165 incidents while the context listed 5.	One test case
Relevance	Answer Relevancy scored 1.00.	One test case
Contextual precision and recall	Both scored 1.00 in DeepEval on the example. RAGAS Context Precision is described as moderate.	One test case; RAGAS value not supplied

8. Limitations

- The incident data is synthetic, so results do not prove performance on real operator data.
- ML metrics come from a held-out split of the same dataset. They show fit to that dataset, not live network accuracy.
- Only one DeepEval test case is reported here, so the 1.00 scores should not be generalised.
- An LLM judge can give different scores between runs and between models. Scores are indicative.
- Retrieval can include loosely related incidents (noise), which lowers context precision.
- The language model works with a limited context window, so only the top-ranked incidents are passed on.
- The historical tests use stored incidents. The system is not connected to live telemetry, so live behaviour is not tested.

9. Recommendations

- Fix the count error behind the 0.86 faithfulness score: take counts from the retrieval result itself, not from the model's wording.
- Add more test cases (different cities, seasons, metrics and recommendation queries) before drawing conclusions.
- Review the Context Precision cases and try different top-k, filter and hybrid-weight settings. Report the new scores as they come out.
- Repeat judge runs on key cases to see how much scores vary.
- Add the saved RAGAS and DeepEval JSON values to this report so every metric has a recorded score.
- Confirm the exact thresholds and timeout/retry values from the code and list them in the next version.

10. Source Files and References

- Technical Stack document (judge provider and model, evaluation tools, report location).

- Project architecture brief (evaluation modules and test types).
- Evaluation results supplied for the ML models (80,000 / 20,000 split), the 0.96 RAG retrieval quality score and the latency_above_130 DeepEval case.
- Not available: the evaluations/reports/ JSON files, threshold settings and frontend qualification logic.